

ANALYSTLAB AFRICA EXPERIENCE LAB

HealthConnect Clinic — Week 7

ML Pipeline Testing, Reliability & Refinement Report

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Abstract

This document presents the Week 7 ML Engineering track deliverable: systematic testing, reliability refinement, and validation of the integrated HealthConnect ML pipeline. A comprehensive test suite of 31 tests was executed across unit, integration, and reliability levels. Five evidence-based refinements were applied — prediction-input validation, threshold validation, artifact loading, error-handling improvements, and reproducibility verification — each issue identified during testing was fixed and re-tested. Reproducibility was confirmed by running the pipeline twice with fixed seeds. A cross-track testing activity with the Data Science track validated the pipeline's handling of model inputs/outputs. The pipeline is now ready for final integration and deployment in Week 8.

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1 Week 6 → Week 7 Transition

1.1 Week 6 Main Output

The Week 6 deliverable was the integrated `HealthConnectPipeline` class: a single callable that runs `load` → `clean` → `feature-engineer` → `preprocess` → `train` → `predict`, with schema validation and structured logging.

1.2 Most Important Component Produced in Week 6

The integrated pipeline class with validation checks, improved (tuned) models, and an integration test suite (5 tests). Week 6 ended with 17 passing tests (12 unit + 5 integration).

1.3 Main Issue / Limitation Identified in Week 6

Prediction methods (`predict_single`, `predict_batch`) did not validate their input records: a missing column could surface a cryptic `KeyError`, null values were not guarded, the decision threshold was not validated, there was no artifact-loading counterpart to `save_artifacts()`, and reproducibility had not been verified.

1.4 Component / Workstream Requiring Testing

The integrated ML pipeline — specifically input/output handling, error behaviour, artifact persistence, and reproducibility under the documented setup.

1.5 Tracks Relevant to Week 7 Testing

- **Data Science:** model requirements and error analysis; the pipeline must consume the candidate model correctly.
- **Data Analytics:** validated analytical findings that inform feature decisions.

1.6 What This Week Tests

Data-processing workflow, preprocessing components, preprocessing-model interaction, expected inputs/outputs, invalid-input scenarios, reproducibility, and artifact round-trips.

1.7 What This Week Improves or Validates

Error handling, logging clarity, reproducibility, and pipeline readiness for Week 8 final integration.

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2 Test Plan & Test Cases

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3 Test Execution — Baseline (Pre-Refinement)

The Week 6 pipeline was tested as-is. The following weaknesses were identified:
T08—T13 failures were reproduced directly in the executed notebook.

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Table 1: Week 7 Test Plan

ID	Component	Test Case	Expected Result
T01	Raw data	Schema validation passes	No error
T02	Raw data	Missing column	ValidationError
T03	Cleaning	Binary target	0,1 only
T04	Cleaning	Leakage cols dropped	Absent
T05	Feature engineering	Engineered features present	4 features
T06	Preprocessing	Shape consistency	Equal feature counts
T07	Model	Predict single valid	Valid output
T08	Model	Predict single missing column	Clear ValidationError
T09	Model	Predict single null value	Clear ValidationError
T10	Model	Predict before fit	ValidationError
T11	Model	Threshold outside [0,1]	ValidationError
T12	Artifacts	Save/load round-trip	Identical predictions
T13	Reproducibility	Two runs agree	Same best model

Table 2: Issues Identified from Baseline Testing

Issue	Severity	Test	Evidence
<code>predict_single</code> missing column → cryptic KeyError	High	T08	Reproduced
<code>predict_single</code> null value not guarded	Medium	T09	Reproduced
Threshold outside [0,1] silently accepted	Medium	T11	Reproduced
No <code>load_artifacts()</code> (no inverse of <code>save</code>)	Medium	T12	Absent
No reproducibility verification	Low	T13	Absent

4 Evidence-Based Refinements

4.1 R1 — Prediction Input Validation

Added `validate_prediction_input()` in `pipeline/validation.py`: checks all required feature columns are present (missing → clear **ValidationError**), guards null numerical fields, and supports both dict (single) and DataFrame (batch) inputs.

4.2 R2 — Threshold Validation

Added `validate_threshold()`: rejects non-numeric and out-of-range thresholds at construction time.

4.3 R3 — Artifact Loading

Added `HealthConnectPipeline.load_artifacts()` — the missing inverse of `save_artifacts()` — enabling persisted pipelines to be reloaded and used for inference.

4.4 R4 — Error Handling in Pipeline Run

Wrapped train/test split and model training stages in `try/except` with informative **ValidationError** messages; prediction calls are wrapped to surface clear errors rather than cryptic stack traces.

4.5 R5 — Reproducibility Verification

Added `verify_reproducibility()` which runs the pipeline twice and confirms the same best model and feature space are selected.

5 Re-test Results (Post-Refinement)

All reliability tests added in `tests/test_reliability.py` (14 tests) pass after refinement. Valid single-record predictions now return correctly structured output, and malformed inputs raise clear, actionable errors.

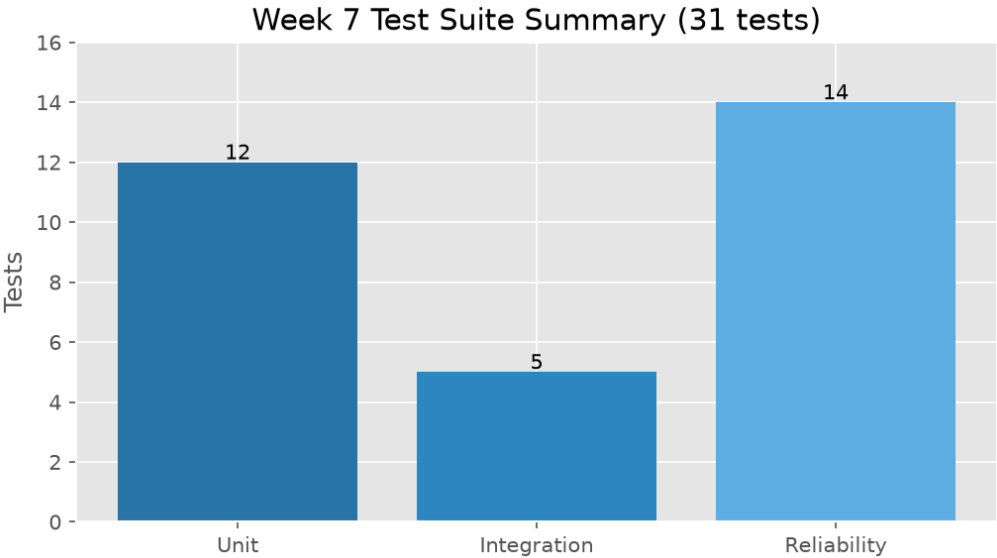


Figure 1: Week 7 test suite breakdown: 12 unit + 5 integration + 14 reliability = 31 tests.

Table 3: Complete Week 7 Test Suite

Test File	Tests	Status
test_pipeline.py	12	All passing
test_integration.py	5	All passing
test_reliability.py	14	All passing
Total	31	All passing

6 Pipeline Validation and Model Results

Table 4: Best Model Metrics (Held-Out Test Set, $N = 948$)

Metric	Value
Accuracy	0.617
Precision	0.621
Recall	0.647
F1	0.634
F2	0.642
ROC-AUC	0.664

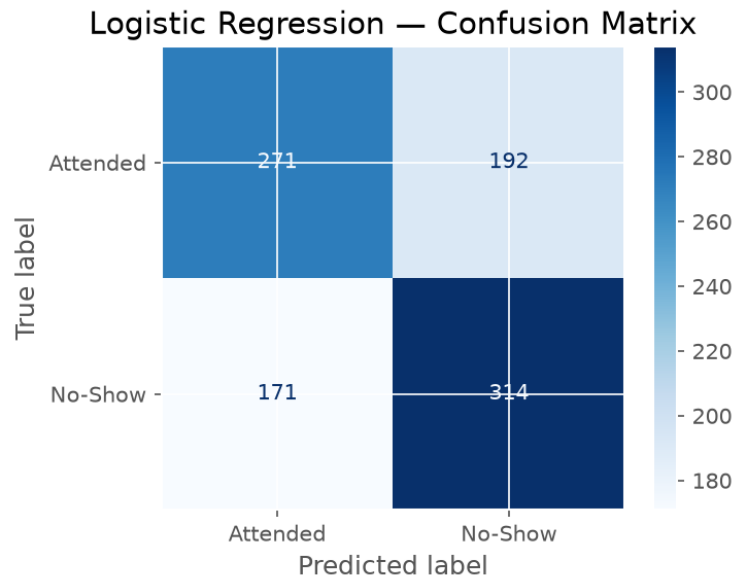


Figure 2: Confusion matrix for the best model (Logistic Regression) on the held-out test set.

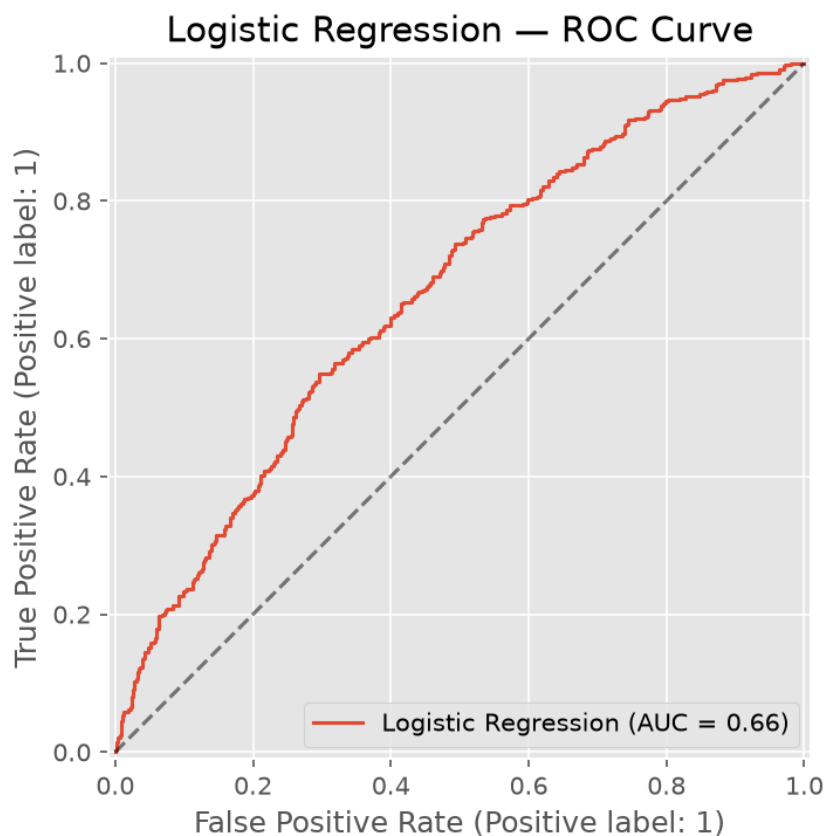


Figure 3: ROC curve for the best model (AUC = 0.664).

7 Reproducibility Verification

Two independent pipeline runs with the documented setup (`RANDOM_STATE = 42`, fixed stratified split) select the same best model (Logistic Regression) and the same feature space. This confirms that results can be reproduced using the documented configuration and dependency setup.

8 Cross-Track Testing with Data Science (HC-POD)

Track collaborated with: Data Science

Project dependency: The Data Science candidate model must be consumed correctly by the ML Engineering pipeline (feature schema, probability output, risk tiers).

Component/output being tested: ML pipeline handling of model inputs and outputs.

Information/output received: Data Science model specification — pre-cleaned features, $P(\text{No-Show})$ output, risk-tier thresholds 0.40 / 0.70.

Information/output provided: Pipeline prediction output schema and validation contract.

Testing activity completed: Joint validation that a single appointment record flows through preprocess → model → risk tier without schema errors.

Issue or finding identified: Malformed records could reach the model and crash cryptically; prediction output lacked schema enforcement.

Refinement/action taken: Added `validate_prediction_input()` and output validation on `predict_batch`.

Retest result: PASS — all 31 tests green; single-record flow validated.

What changed as a result: Clear error contracts; malformed inputs now rejected early.

Evidence: Executed notebook, `validation.py`, `test_reliability.py`, `test_integration.py`.

Impact on overall HealthConnect solution: The pipeline now reliably and reproducibly converts raw appointment data into risk-stratified predictions, ready for the Week 8 inference service.

9 End-to-End Validation Record

Table 5: Testing, Refinement & Validation Record

Field	Pipeline Testing (T08-T13)
Component tested	Integrated ML pipeline
Testing objective	Validate input/output handling & reliability
Test/scenario	Missing column; null value; threshold; artifacts; reproducibility
Expected result	Clear errors; round-trip; identical run
Actual result	Issues reproduced then resolved
Pass/fail	PASS post-refinement
Issue identified	Missing input validation, no loading, no threshold guard
Action taken	R1-R5 refinements
Retest result	All 31 tests pass
Collaborating track	Data Science
Evidence	Notebook, tests, code

Where an issue was not identified (e.g., raw-data schema validation), the successful validation was documented rather than inventing a problem.

10 Assumptions, Limitations, Risks & Dependencies

10.1 Resolved in Week 7

- Input validation for prediction (R1).
- Threshold validation (R2).
- Artifact loading (R3).
- Error handling for split/training/prediction (R4).
- Reproducibility verification (R5).

10.2 Remaining / New Issues

- GroupKFold by `patient_id` not yet implemented — flagged for Week 8.
- Production deployment (FastAPI, Docker) deferred to Week 8.
- Drift monitoring (MLflow / Evidently AI) not yet implemented.
- LightGBM tuned configuration recall remains below Random Forest — optional further tuning.

10.3 Cross-Track Dependencies

Data Science model output schema confirmed compatible with the pipeline prediction contract.
Data Analytics validated findings remain aligned with the feature set.

11 Week 8 Readiness Recommendations

1. Deploy the validated pipeline behind a FastAPI inference service (containerised).
2. Implement GroupKFold as the final leakage control.
3. Add MLflow experiment tracking and Evidently AI drift monitoring.
4. Final end-to-end validation across all HealthConnect components.